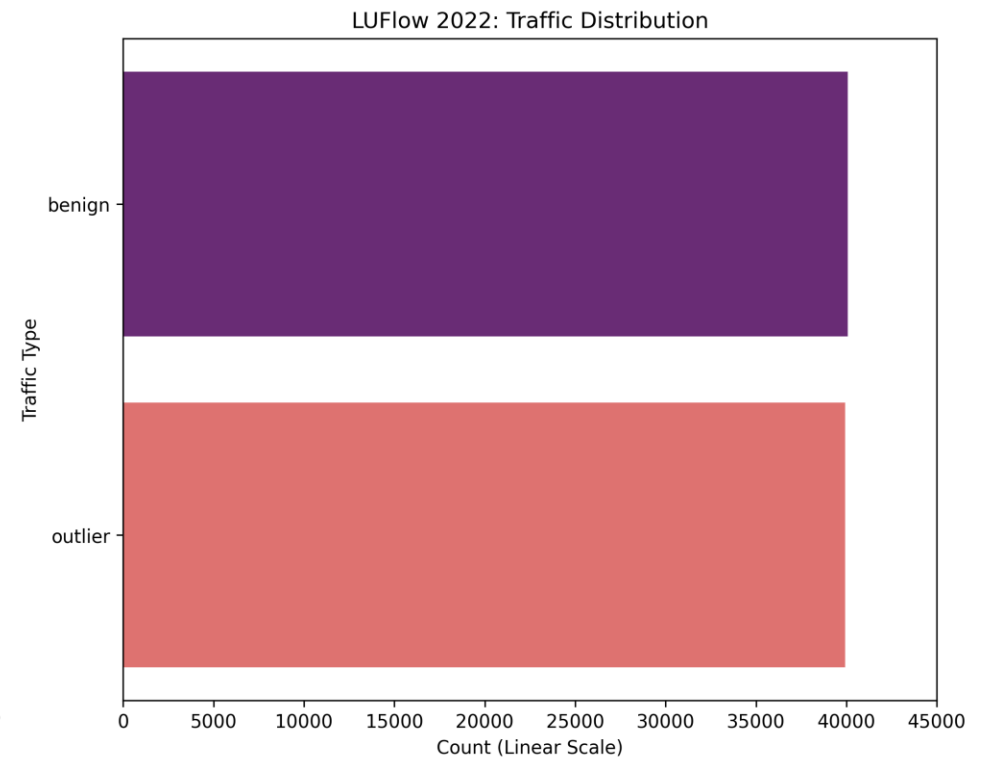
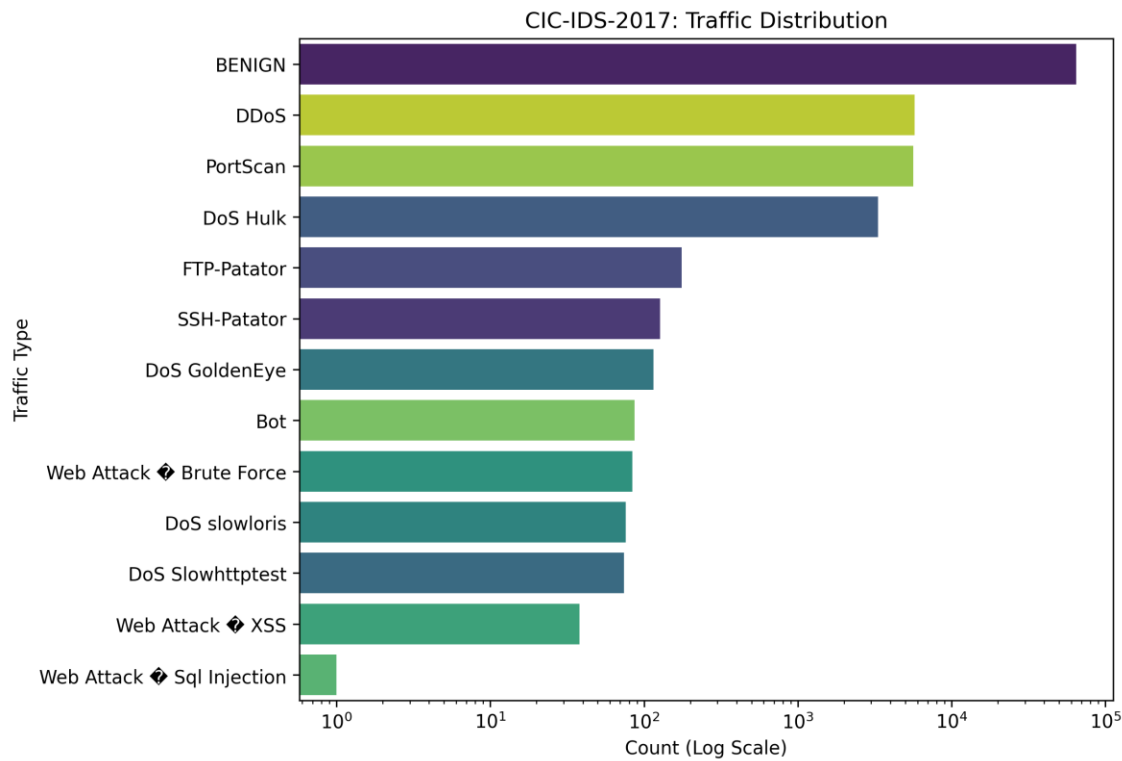


Unsupervised Machine Learning for Network Traffic Clustering

A Comparative Analysis of Lab vs. Real-World Telemetry

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- To test this, I compared two very different datasets. On the left, we have a lab dataset from 2017, on the right, we have a real-world dataset from 2022. Notice the difference: the real internet is so chaotic that it's practically a 50/50 split of normal and malicious traffic.

The Lab vs. The Real World

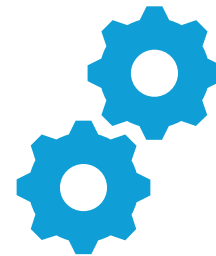
Data Cleaning & Processing



Cleaned: Removed errors like infinite flow durations.



Standardized: Scaled all data so features like 'packet size' didn't mathematically overpower 'duration'.



Optimized: Ran the algorithm on a 20,000-record subset to prevent memory overflow.

Why DBSCAN?

Stands for: Density-Based Spatial Clustering of Applications with Noise.

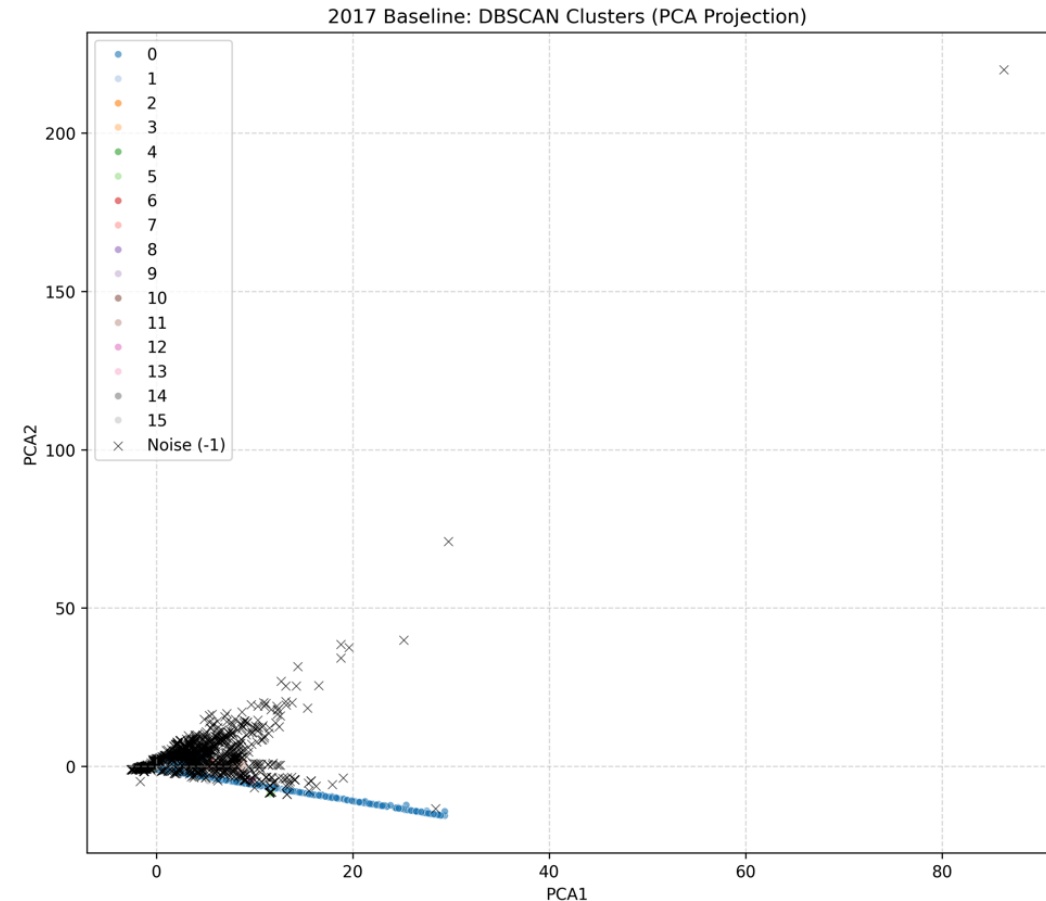
Groups tight, dense packets of data together.

Officially flags isolated, weird data points as "Noise" (Anomalies).

2017 Results (The Sterile Lab)

Found 16 highly specific micro-clusters.

Flagged 5.47% of traffic as noise.



2022 Results (The Chaos)

Found only 7 massive macro-clusters.

Flagged just 1.61% of traffic as noise.



The Mathematical Paradox



2017 Lab Score: **0.29**



2022 Real-World Score: **0.37**

Conclusion

Unsupervised clustering easily catches basic, scripted attacks in controlled environments.

Modern internet background noise severely degrades the algorithm's performance.

Future defenses need hybrid models to handle the complexity of today's cyber threats.